ENVIROLEARN

A Gamified Environmental Education Platform for Schools

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INDEX

- Introduction
- Problem Statement
- Literature Review
- Gaps Identified
- Proposed Solution
- System Architecture
- Use Case Diagram
- Data Flow Diagram
- Methodology
- Functional Requirements
- Non Functional Requirements
- Software and Hardware Requirements
- Results
- Conclusion
- Future Scope
- References

INTRODUCTION

- Environmental issues like climate change, pollution, and resource depletion are increasing global challenges.
- Early environmental education is essential to develop responsible and sustainability-oriented citizens.
- Schools play a vital role in shaping students' attitudes toward environmental conservation.
- Technological advancements have made digital learning platforms effective for improving student engagement.
- Gamification enhances education by making learning interactive, motivating, and memorable.

PROBLEM STATEMENT

To develop a software to solve the problem of traditional, low-interactivity environmental education in schools, which leads to reduced student engagement, awareness, and participation in sustainability practices.

LITERATURE REVIEW

Name of Paper	Year	Methodology	Result	Limitation
Gamified Learning Platforms: Transforming Student Engagement through Digital Tools	2025	Mixed-method study using surveys, platform analytics, and case analysis. Examines the use of gamified digital tools such as points, badges, leaderboards, and interactive challenges across school and higher education settings.	Gamified learning platforms significantly improve student engagement, motivation, and academic performance. Students showed higher participation and improved learning outcomes due to real-time feedback and interactive elements.	Over-reliance on extrinsic rewards may reduce intrinsic motivation over time. Effective results depend on thoughtful design, access to technology, and proper alignment with educational goals.

LITERATURE REVIEW

Name of Paper	Year	Methodology	Result	Limitation
Online Environmental Gamification and University Students' Pro-Environmental Organizational Citizenship Behaviour	2025	Survey-based study using online environmental gamification platforms (e.g., Ant Forest). Data collected from university students and analysed using regression and mediation analysis based on CAPS theory.	Online environmental gamification significantly increases students' pro-environmental behaviour. Environmental responsibility and environmental passion act as mediators in promoting sustainable actions.	Study focused only on university students and relied on self-reported survey data. Results may not generalize to school-level education or younger students.

LITERATURE REVIEW

Name of Paper	Year	Methodology	Result	Limitation
Gamification of Sustainability Communication: Engaging Audiences Through Interactive Platforms	2020	Mixed-method approach using case studies, surveys, and qualitative analysis. Examines gamified sustainability platforms using game elements such as points, badges, leaderboards, and interactive quizzes.	Gamification improves audience engagement, knowledge retention, and motivation toward sustainable practices. Interactive features and feedback increase participation compared to traditional communication methods.	The study focuses mainly on communication and awareness rather than structured learning outcomes. Does not specifically address school-level implementation or curriculum-based educational platforms.

LITERATURE REVIEW

Name of Paper	Year	Methodology	Result	Limitation
The Game with Impact: Gamification in Environmental Education and Entrepreneurship	2015	Case study-based research conducted by the Moldovan Environmental Governance Academy (MEGA). Analyzes gamified environmental education projects (G.R.E.E.N. and Game with Impact) using game elements such as missions, levels, rewards, and collaborative challenges.	Gamification significantly increased youth engagement, participation, and practical involvement in environmental activities. Participants showed improved environmental awareness, leadership skills, and motivation.	The study is limited to specific regional case studies and small participant groups. Long-term behavior change and scalability of gamified education platforms require further research.

GAPS IDENTIFIED

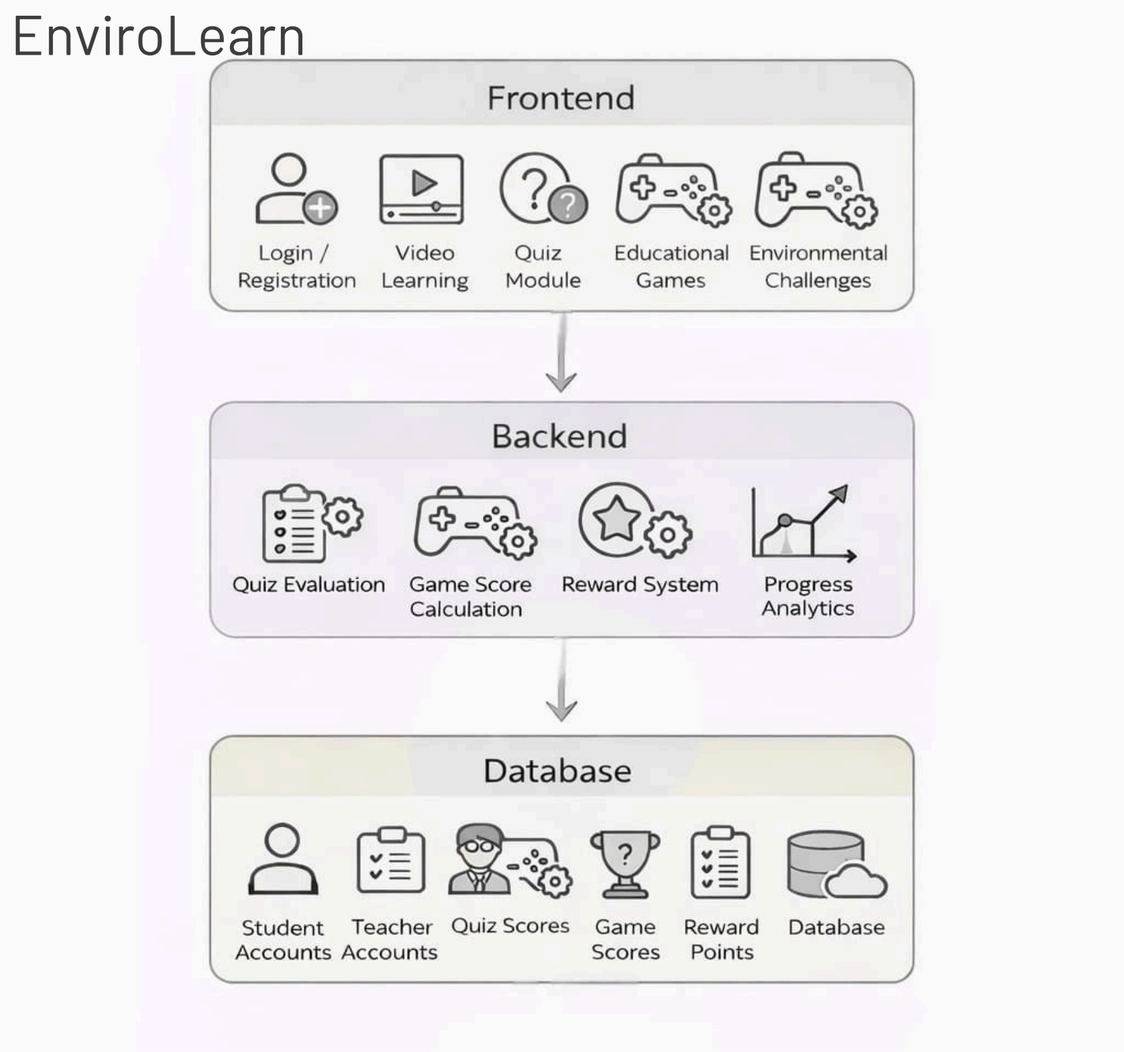
- Most gamification-based environmental education studies focus on university students or general audiences, with limited emphasis on school-level learners.
- Existing research is often restricted to specific regions or small-scale case studies, which limits its broader applicability.
- Many current platforms focus mainly on raising awareness and motivation, but they lack proper alignment with school curricula.
- There is limited support for tracking student progress, performance analytics, and long-term behavioral change, indicating the need for a more scalable and structured gamified platform for schools.

PROPOSED SOLUTION

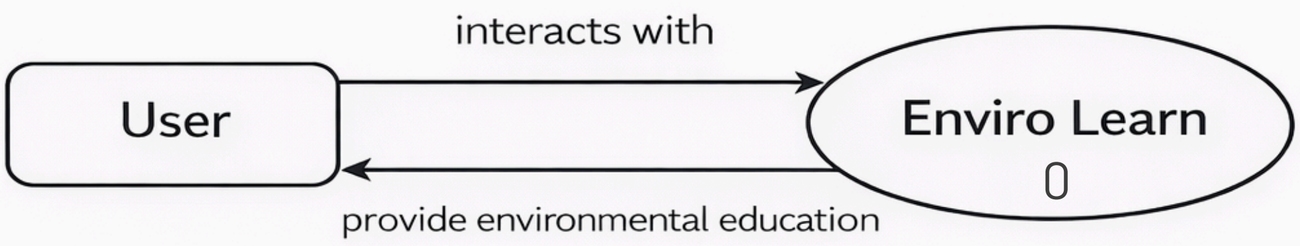
We propose a Gamified Environmental Educational Platform that uses videos, quizzes, and interactive games to educate students about environmental issues and motivate eco-friendly behavior.
The platform combines:

- Environmental learning content
- Gamification techniques
- Students watch short educational videos on environmental topics.
- Students attend quizzes based on video content.
- Interactive games reinforce learning concepts.
- Users earn points, badges, and levels leaderboards, and missions to increase engagement, environmental responsibility, and passion among students.
- Progress is tracked and displayed.

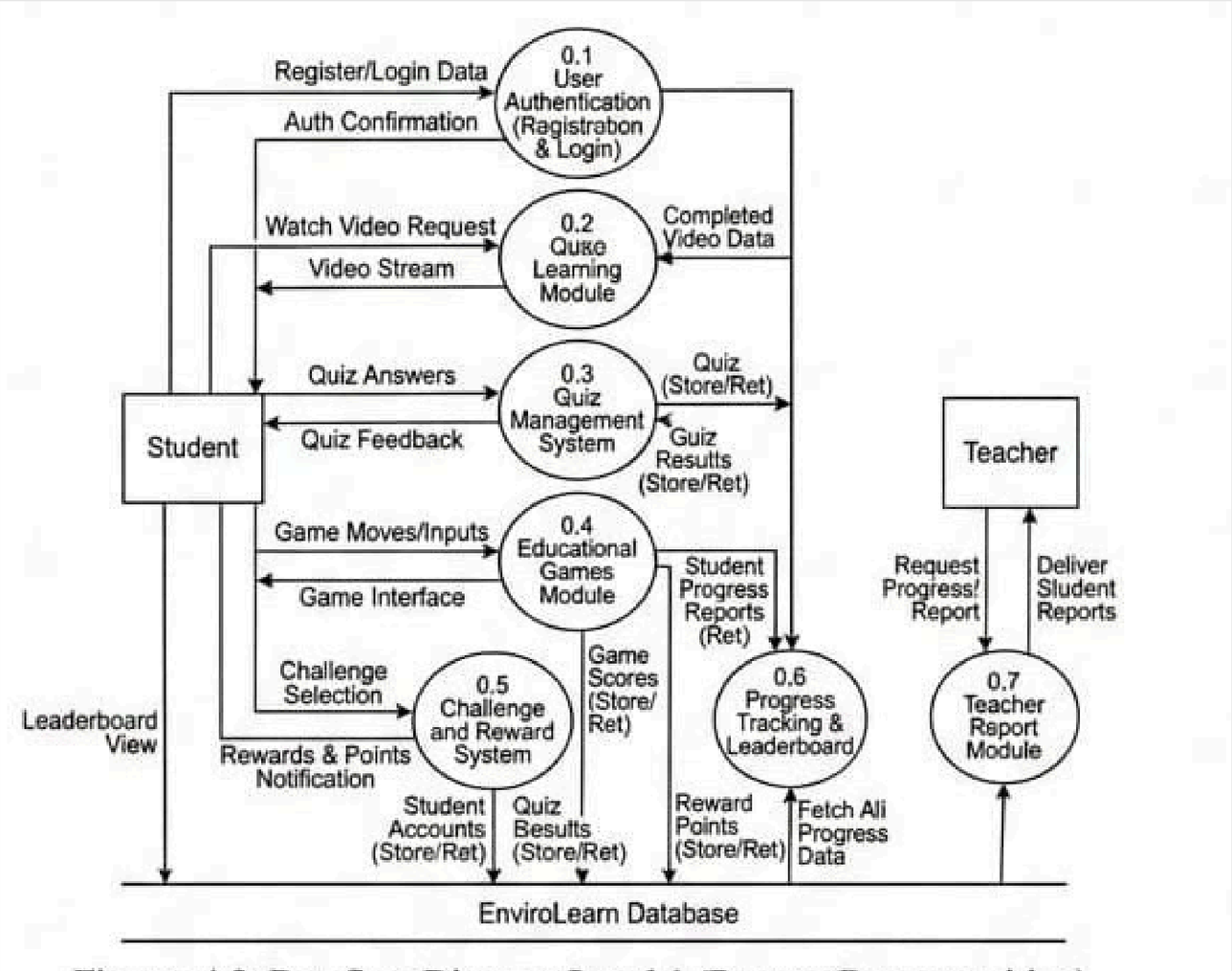
SYSTEM ARCHITECTURE



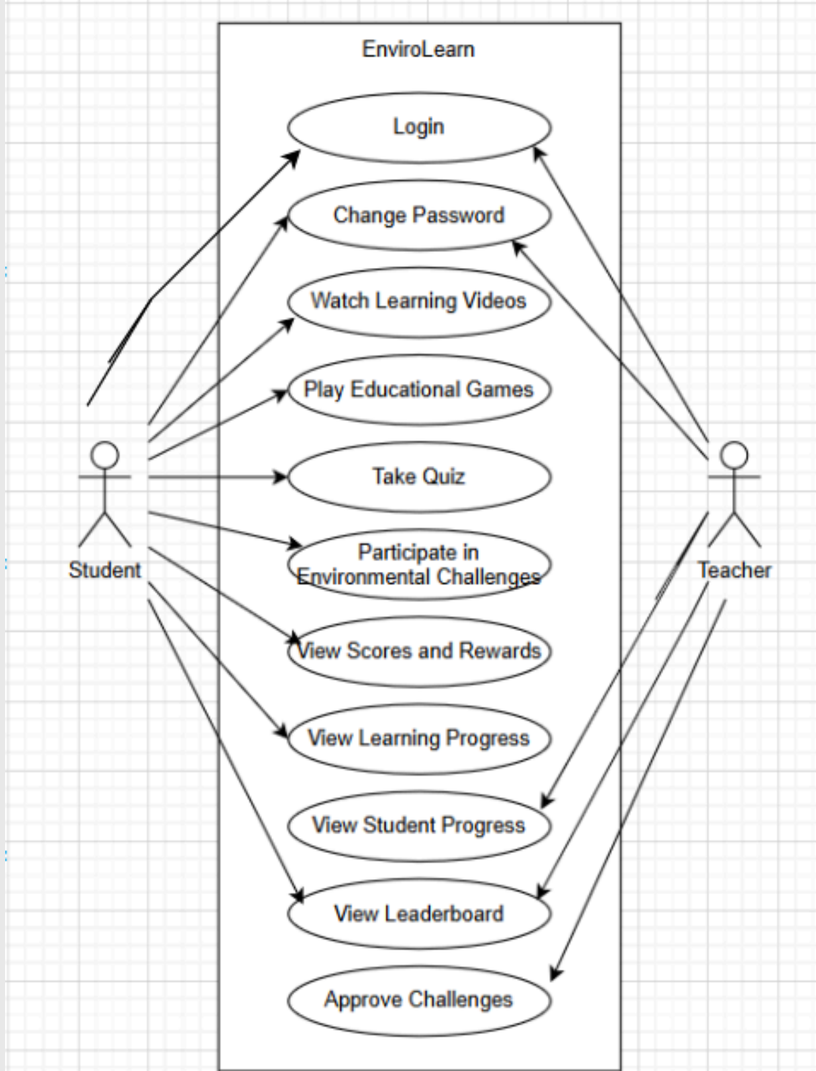
DATA FLOW DIAGRAM: LEVEL 0



DATA FLOW DIAGRAM: LEVEL 1



USE CASE DIAGRAM



METHODOLOGY

1. User Authentication

Technical Method / Algorithm: Session-Based Authentication with Password Hashing

Description: EnviroLearn uses secure authentication mechanisms to verify users during login. Passwords are stored using hashing techniques, and authenticated sessions are created to allow secure access to platform features.

2. Role-Based Access Control

Technical Method: Role-Based Access Control (RBAC)

Description: Users are categorized as Students or Teachers, and access permissions are controlled accordingly. Students can complete various environmental tasks, while teachers can monitor submissions through the administrative dashboard.

METHODOLOGY(CONTD.)

3. Geo-Tagging Verification

Technical Method / Algorithm: Browser Geolocation API Integration

Description: EnviroLearn retrieves the user's live geographic coordinates using the browser's geolocation functionality. This helps verify that environmental activities are performed at the user's actual location.

4. Image Metadata Processing

Technical Method: EXIF Metadata Extraction

Description: Uploaded images are analyzed to extract embedded metadata such as location coordinates and timestamps. This information is used to validate the authenticity of challenge submissions.

METHODOLOGY(CONTD.)

5. Teacher Monitoring Dashboard

Technical Method: Data Aggregation

Description: Teachers access a dashboard that displays student submissions, activity records, and verification details. This enables teachers to monitor participation and assess environmental engagement.

FUNCTIONAL REQUIREMENTS

1. User Management

- The system shall allow students and teachers to signup and log in.
- The system shall authenticate users using username and password.
- The system shall allow users to log out securely.
- The system shall support role-based access (Student / Teacher)

2. Gamification Features

- The system shall award points for correct answers and task completion.
- The system shall assign badges and achievements based on performance.
- The system shall track daily challenges.

FUNCTIONAL REQUIREMENTS(CONTD.)

3. Gamified Video-Based Learning Module

- The system shall allow the platform to provide environmental education video lessons to students.
- The system shall ensure that students watch the complete video session before proceeding further.
- The system shall present gamified questions related to the video content at the end of each session.
- The system shall evaluate student responses and award points based on accuracy and completion.

FUNCTIONAL REQUIREMENTS(CONTD.)

4. Progress Tracking

- The system shall store student progress and scores.
- The system shall allow students to view their progress in the leaderboard.
- The system shall generate performance summaries for students.

5. Teacher Dashboard

- The system shall allow teachers to view student performance reports.
- The system shall provide analytics such as scores, participation, and completion rate.
- The system shall allow teachers to assign games and quizzes to students.

NON FUNCTIONAL REQUIREMENTS

- Performance: fast response for games and quizzes
- Security: secure login and data protection
- Scalability: supports increasing users and modules
- Reliability: stable system with backup support
- Usability: simple, interactive, and user-friendly UI
- Maintainability: modular flask-based architecture
- Availability: internet required for hosting and notifications

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

Frontend: HTML,CSS,Javascript

Backend: Flask(Python)

Database: SQLlite,SQLAlchemy

Hardware Requirements

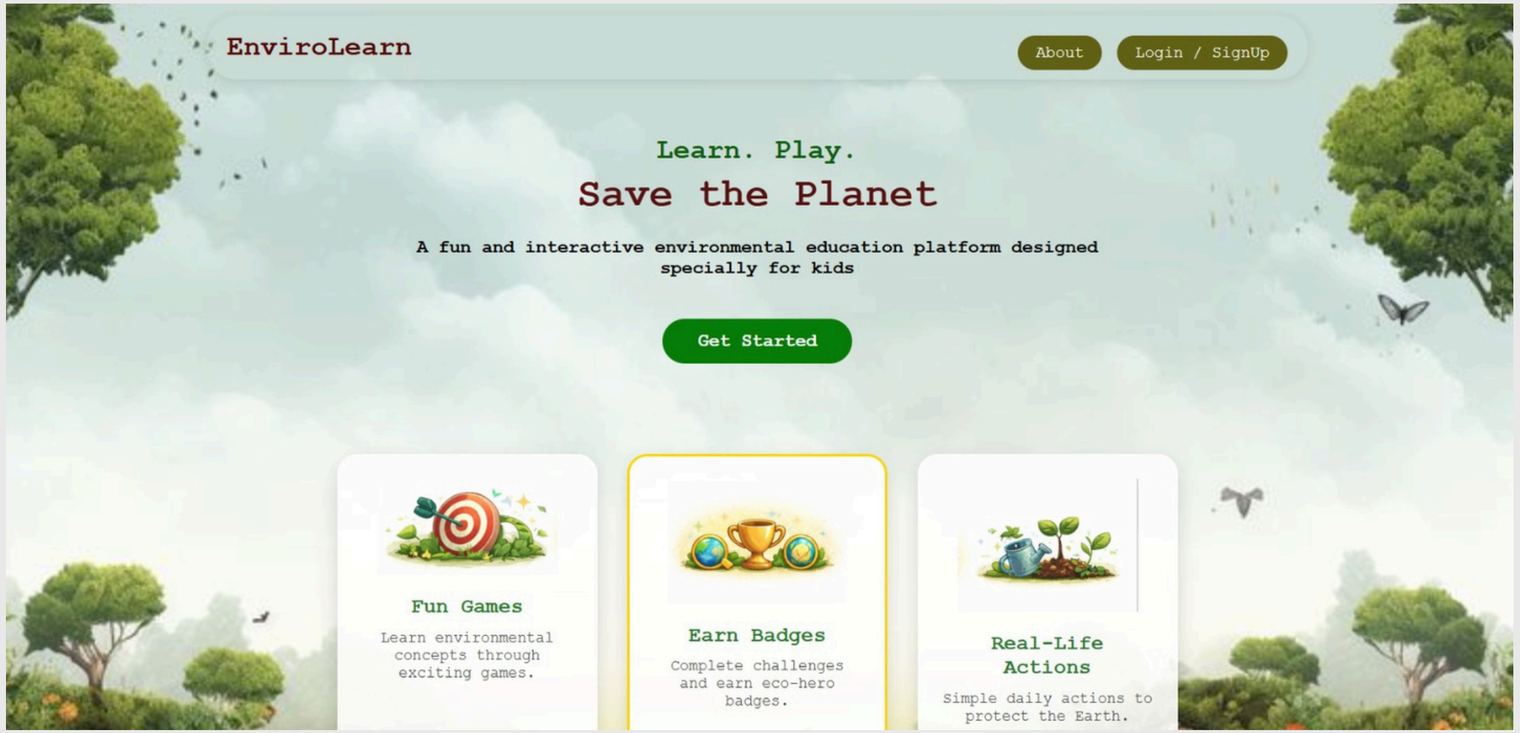
Processor- Intel i3 or above

RAM- Minimum 4 GB

Device- Laptop, Mobile Phone

Internet Connection- Required for hosting and notifications

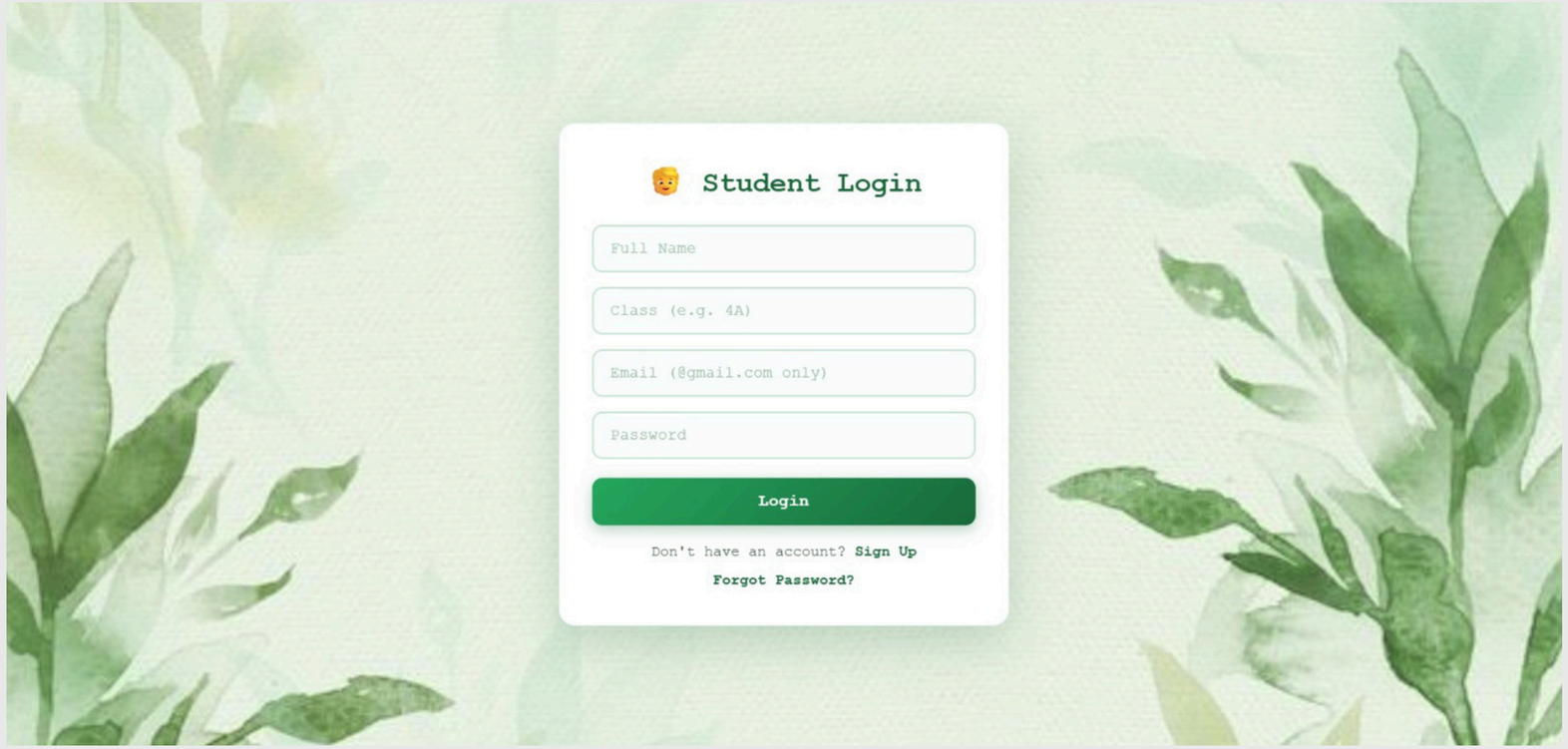
RESULTS



Home Page / Main Dashboard

Main interface where students can access Games, Videos, Quizzes, Challenges, and the Leaderboard.

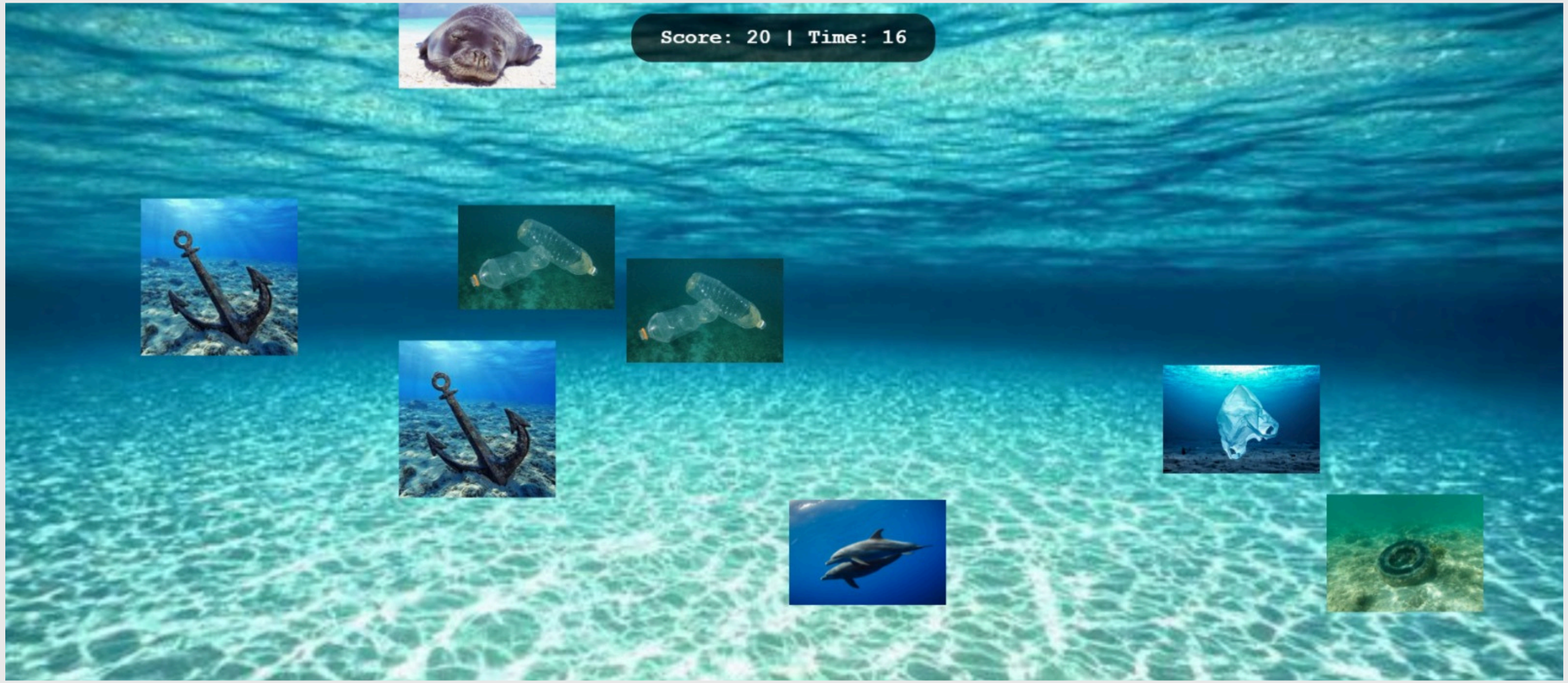
RESULTS



Student Login Page

Login page where students enter their details to securely access the learning platform

RESULTS



Ocean Cleanup Game

Interactive game where students collect ocean waste and learn about marine pollution

RESULTS

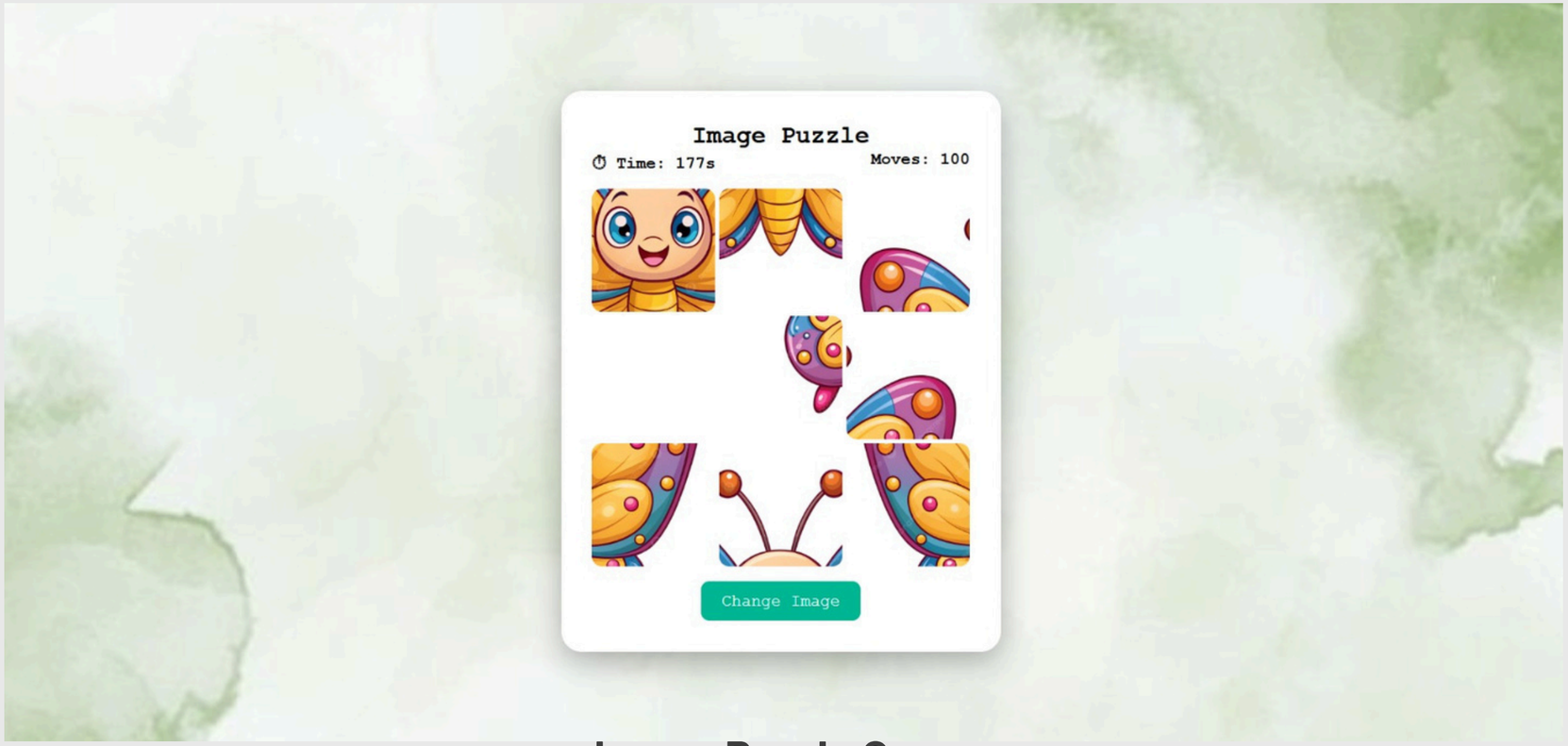


Image Puzzle Game

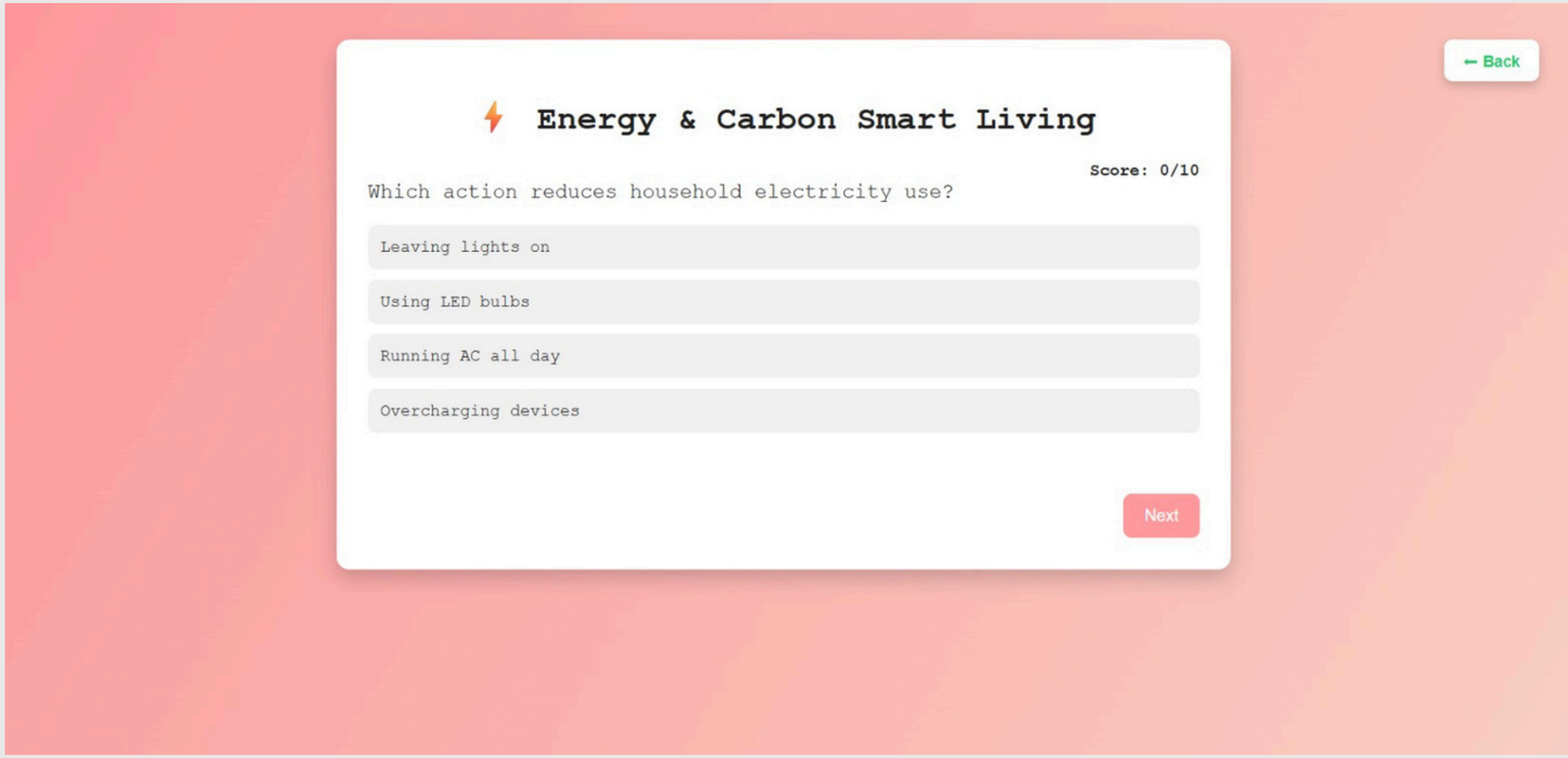
Puzzle-based activity that helps students improve problem-solving skills while learning environmental concepts.

RESULTS

Video based Learning Module

Educational video module that teaches environmental concepts and unlocks a related quiz after watching.

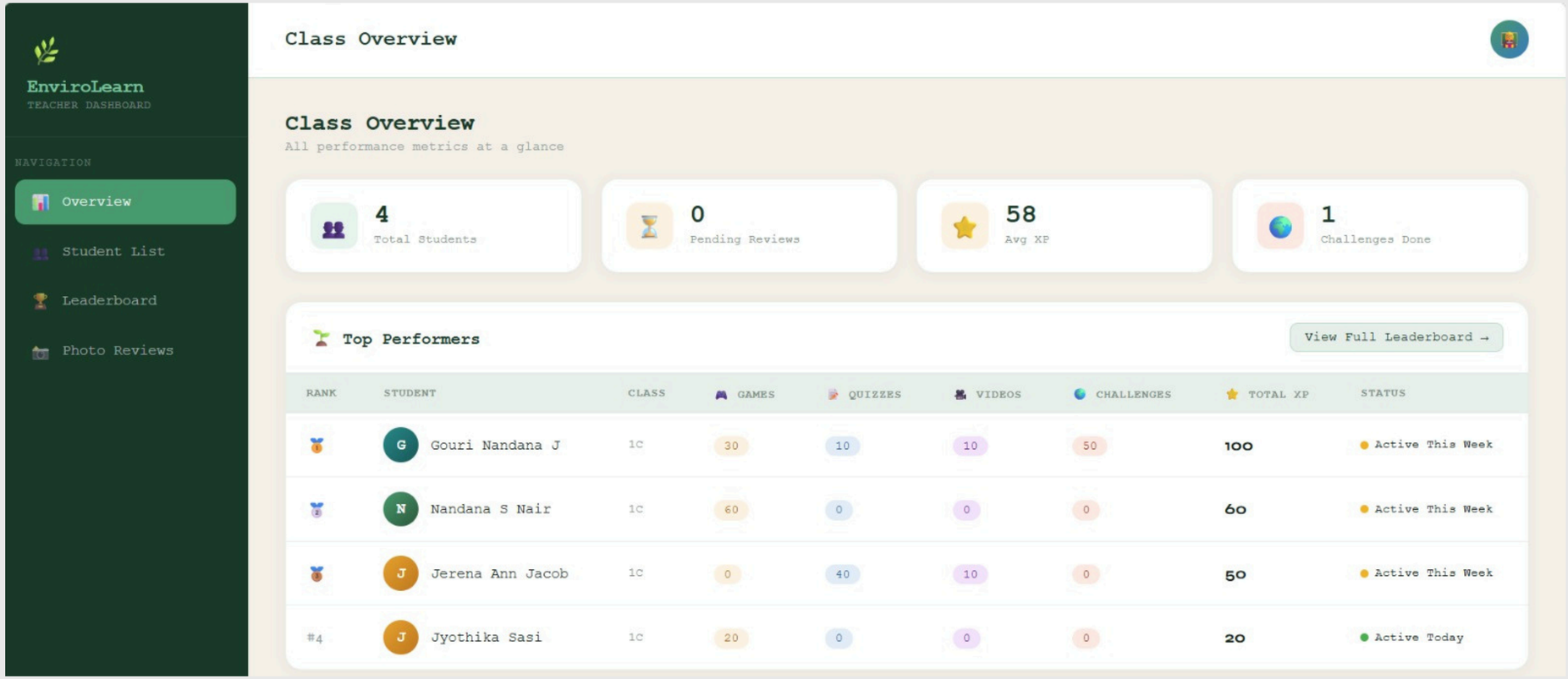
RESULTS



Quiz Module

Interactive multiple-choice quiz designed to test students' knowledge about environmental topics

RESULTS



Teacher Dashboard

Dashboard that displays class overview, student progress, and leaderboard rankings

RESULTS

← Back

Leaderboard

RANK	NAME	CLASS	GAME XP	QUIZ XP	VIDEO XP	CHALLENGE XP	TOTAL XP
1	Gouri Nandana J	1C	300	49	30		429
2	Joshua George	1B	240	30	10		280
3	Nandana S Nair	1C	60	0	0		60
4	Albin Thomas	1B	45	0	10		55
5	Jerena Ann Jacob	1C	0	40	10		50
6	Nayana J Pillai	2A	0	20	10		30
7	Jyothika Sasi	1C	20	0	0		20
8	Fidha Safar	1B	0	6	10		16
9	Raisha Hashly	1B	15	0	0		15
10	Nilofar Fathima	2A	0	0	0		0

Leaderboard

CONCLUSION

- EnviroLearn uses gamification to make environmental education interactive and engaging for school students.
- The platform combines videos, quizzes, and challenges to improve awareness about issues like pollution, climate change, and biodiversity.
- Built with HTML, CSS, JavaScript, and Python Flask, the system provides a simple and effective digital tool for sustainability education.

FUTURE SCOPE

The EnviroLearn platform can be further improved with advanced features to increase its effectiveness and scalability. Possible future enhancements include:

- AI integration to personalize learning paths based on student performance.
- More environmental games, simulations, and interactive activities to make learning more engaging.
- Mobile application development for better accessibility and wider reach.
- Real-time leaderboards and collaborative eco-challenges between schools.
- Integration with school curriculum standards to support formal environmental education.
- Analytics dashboards for teachers to monitor student progress and awareness levels.

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Thank you